

Milestone 2: Analytics Implementation – Model Building & Evaluation

Arrest Prediction for Chicago Crime Data Using Spatial, Temporal, and Crime-Type Features

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IT5006 Fundamentals Of Data Analytics

AY2025/2026 Semester 2

Submission Date: 15 March 2026

1. Problem Definition and Data Preparation

1.1 Problem statement and analytical approach

This project develops a **binary classification model** to predict whether a recorded crime incident in the Chicago dataset results in an arrest ($Arrest = 1$). Only variables related to **location**, **time**, and **crime type** are used as predictors.

The study is **predictive rather than causal**. Its purpose is to identify recurring patterns associated with arrest outcomes, not to explain why an arrest occurred in any individual case. The analytical approach follows a standard supervised learning pipeline: merge multi-year files, clean and transform raw variables, engineer compact predictors, train several classification models, compare them with multiple evaluation metrics, tune the best candidate, and interpret the final model for practical use.

1.2 Data source, preprocessing, and cleaning

The dataset combines annual Chicago crime training files from **2015 to 2025**, producing **2,755,736 records** and **22 original variables**.

Table 1. Data overview

Item	Value
Data source	Chicago crime training files (2015–2025)
Total records	2,755,736
Original variables	22
Prediction target	Arrest
Task type	Binary classification

All annual files were concatenated into one dataset. The Date variable was converted into datetime format so that temporal fields could be extracted. Location Description was standardized into **10 broader location categories** and encoded as UNIFIED_LOCATION_CODE. Primary Type was remapped into two crime-related variables: *crime_against* and *offense_category_name*.

A small number of observations (**6,605 records**) were not fully matched by the crime-mapping dictionary. These cases were retained to avoid unnecessary data loss, but future work should extend the mapping rules or introduce an explicit **Unknown** class to improve consistency.

1.3 Feature engineering strategy

Feature engineering was designed to keep the model both **predictive** and **interpretable**.

- **Spatial feature:** UNIFIED_LOCATION_CODE
- **Temporal features:** *month, day, hour, weekday, is_weekend, is_holiday*
- **Crime-type features:** *crime_against, offense_category_name*

Categorical variables were one-hot encoded before modeling. The final feature matrix contained **33 variables**. Data were split using an **80/20 stratified train-test split** so that the arrest proportion remained similar across training and test sets.

2. Model Implementation and Training

2.1 Implemented models

Four models were implemented to balance predictive strength, interpretability, and methodological diversity:

1. **Logistic Regression** – a linear probabilistic baseline
2. **Decision Tree** – an interpretable nonlinear classifier
3. **Random Forest** – a bagging-based ensemble of trees
4. **Gradient Boosting** – a boosting-based ensemble that improves weak learners sequentially

This set satisfies the requirement to implement multiple models while allowing comparison between a simple linear benchmark and more flexible nonlinear approaches.

2.2 Model architectures and training process

All models were trained on the same engineered feature set and evaluated on the same held-out test set for fair comparison. Logistic Regression provides a baseline for linearly separable structure. The Decision Tree captures nonlinear interactions and remains easy to interpret. Random Forest reduces variance through averaging across many trees, while Gradient Boosting improves predictive fit by sequentially correcting previous errors.

Because the target classes are imbalanced, the training process emphasized metrics beyond accuracy alone. Model comparison therefore focused on **Precision, Recall, F1-score, and AUC-ROC**, in addition to Accuracy.

2.3 Hyperparameter tuning methodology

The **Decision Tree** was selected for additional tuning because it performed competitively while remaining interpretable. Hyperparameter tuning was conducted using **GridSearchCV** on a stratified sample of **50,000 training observations**, with

F1-score as the optimization metric because it balances precision and recall under class imbalance.

The search focused on tree-complexity controls such as maximum depth, minimum samples split, and minimum samples leaf, while class balancing was fixed through *class_weight='balanced'*. The final tuned tree used moderate complexity constraints, which reduced overfitting while preserving useful recall. Although the tuned model did not materially outperform the original balanced tree on the final test set, the tuning process confirmed that a moderately constrained tree was an appropriate specification.

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3. Model Evaluation and Comparison

3.1 Standard classification metrics

Since this is an imbalanced classification problem, **accuracy alone is not sufficient**. A model can achieve high accuracy by favoring the majority class while still performing poorly on arrest cases. For that reason, this report evaluates models using **Accuracy, Precision, Recall, F1-score, and AUC-ROC**.

- **Precision** measures how reliable positive arrest predictions are.
- **Recall** measures how many true arrest cases are identified.
- **F1-score** balances Precision and Recall.
- **AUC-ROC** measures the model's overall ranking and class-separation ability.

Table 2. Model Performance Summary

Model	Accuracy	Precision (Arrest=1)	Recall (Arrest=1)	F1-score	AUC-ROC
Logistic Regression	0.77	0.40	0.65	0.49	0.806
Decision Tree	0.82	0.49	0.67	0.57	0.852
Random Forest	0.82	0.49	0.66	0.57	0.849
Gradient Boosting	0.88	0.83	0.42	0.56	0.847

Tree-based models outperform Logistic Regression, showing that the relationship between arrest outcomes and the predictors is not purely linear. Although Gradient Boosting achieves the highest accuracy and precision, its recall is much lower. The **Decision Tree** was therefore selected as the final model because it provides the best balance among **Recall, F1-score, AUC-ROC, and interpretability**.

3.2 Spatial accuracy evaluation

Spatial performance was evaluated by location category to test whether predictive quality varies across different environments.

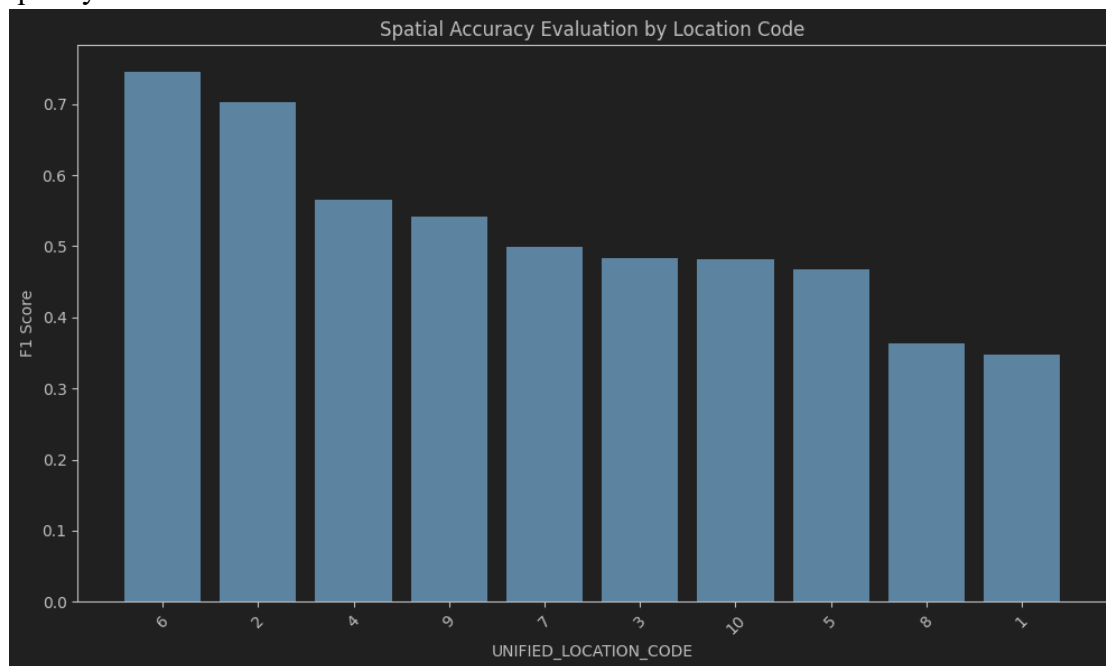


Figure 1. Spatial performance by location category

Note. The figure shows F1-scores across unified location categories.

Table 3. F1-score by location category

Code	Category	Count	F1
6	Government/Public	571	0.745
2	Street/Outdoor	190,388	0.703
4	Retail/Commercial	20,121	0.566
9	Parking Lot	9,349	0.542
7	Medical Facility	1,594	0.499
3	Transportation Hub	7,079	0.483
10	Unknown/Other	220,226	0.482
5	Entertainment	4,069	0.467
8	Workplace/Office	5,562	0.364
1	Residence	92,189	0.348

The results indicate substantial spatial heterogeneity. Performance is strongest in **Street/Outdoor** and **Government/Public** settings, although the latter has a very small sample and should be interpreted cautiously. Performance is weakest in **Residence**, suggesting that residential incidents may depend on contextual factors not captured by the current features.

3.3 Temporal accuracy measures

Temporal evaluation was conducted at two levels: **hour of day** and **weekday versus weekend**.

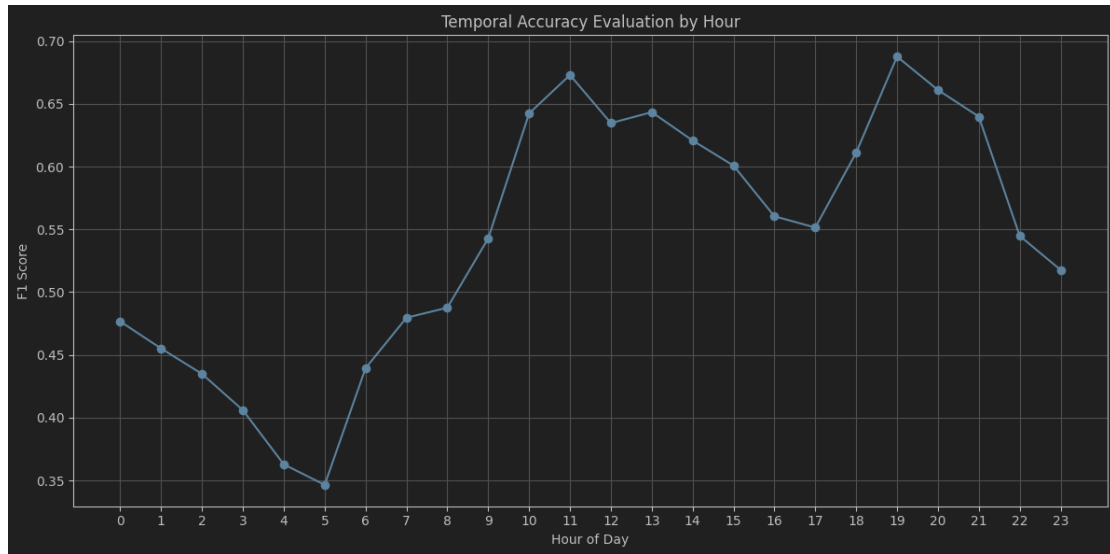


Figure 2. Hourly model performance

Note. The figure reports hourly F1-scores of the final Decision Tree model.

Hourly performance is weakest in the **early morning**, especially around **04:00–05:00**, improves after **09:00**, remains relatively strong from **10:00 to 15:00**, and rises again during **18:00–21:00**. The highest hourly value occurs at **19:00**, with **F1 $\approx$ 0.688**. This suggests that **hour of day is the most informative temporal predictor**.

Table 4. Weekday and weekend performance

Day Type	Count	Accuracy	Precision	Recall	F1
Weekday	393,521	0.841	0.535	0.650	0.587
Weekend	157,627	0.776	0.418	0.707	0.525

Weekdays show better balanced performance, while weekends produce higher recall but lower precision. In practice, this means weekend predictions capture more arrest cases but also generate more false positives.

3.4 Model robustness analysis and cross-validation

Robustness was examined in two ways: **cross-validation stability** and **sensitivity to class balancing**.

A **5-fold cross-validation** was conducted on a stratified sample of **50,000 observations** for the balanced Decision Tree.

Table 5. Five-fold cross-validation results

Metric	Mean
Accuracy	0.811
Precision	0.469
Recall	0.652
F1-score	0.545
AUC-ROC	0.819

The relatively small variation across folds indicates that model performance is reasonably stable and not driven by a single train-test split.

An additional robustness check removed class balancing from the Decision Tree. The **unbalanced** tree achieved **Accuracy = 0.89**, **Precision = 0.83**, **Recall = 0.45**, **F1 = 0.58**, and **AUC = 0.852**. This shows that performance is highly sensitive to class-imbalance handling: the unbalanced model becomes more conservative, greatly improving precision but missing many true arrest cases. The balanced version is preferred because it provides a more useful precision-recall trade-off for this task.

4. Model Interpretation and Business Insights

4.1 Feature importance analysis

Feature importance was examined using the balanced Decision Tree specification adopted for interpretation.

Table 6. Top feature importances

Rank	Feature	Importance
1	crime_against_Society	0.565
2	UNIFIED_LOCATION_CODE	0.166
3	crime_against_Person	0.110
4	offense_category_name_Other Offenses	0.093
5	hour	0.015

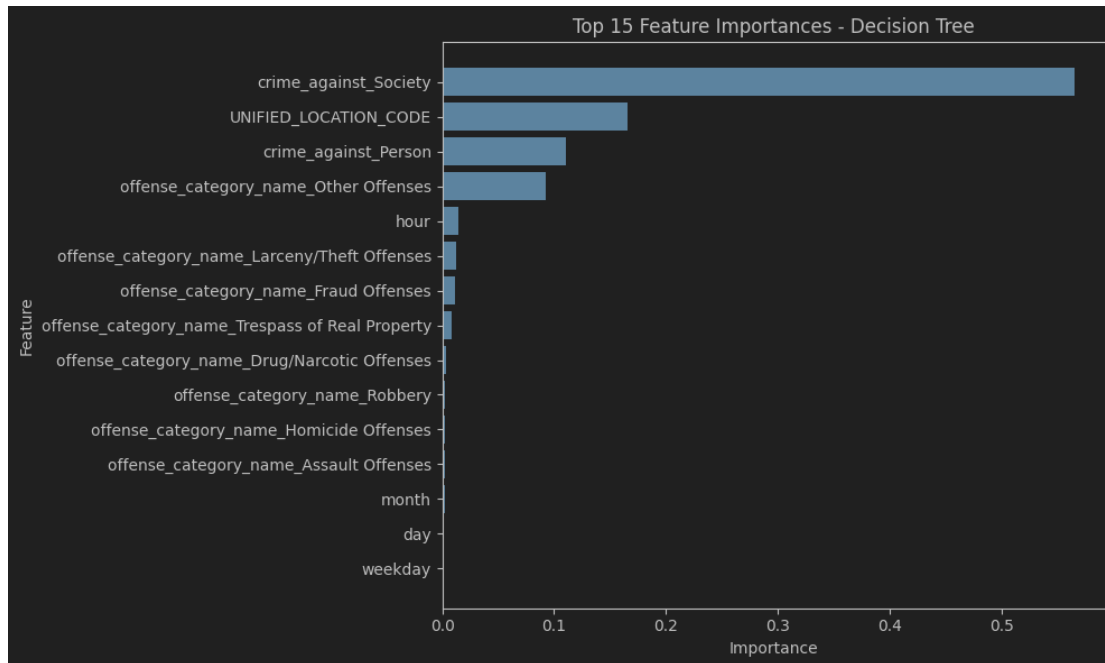


Figure 3. Top 15 feature importances of the Decision Tree

Note. Feature importance values are derived from the final balanced Decision Tree model.

The model relies primarily on **crime type**, followed by **location type** and **hour of day**. This means arrest predictability is shaped most strongly by **what happened**, **where it happened**, and **when it happened**, with crime type clearly the dominant predictor.

4.2 Actionable insights for law enforcement

The model provides several practical insights for law-enforcement planning, while remaining a decision-support tool rather than an automated decision-maker.

First, **crime type should be the starting point for operational screening**, because it contributes the most predictive value. Second, resources may be used more efficiently when combined with **location-sensitive interpretation**: the model performs better in **Street/Outdoor** contexts than in **Residence**, so prediction confidence should not be treated as uniform across places. Third, the strong hourly pattern suggests that **time-aware deployment** may be useful; predictive performance is generally stronger during daytime and early evening than in the early morning. Fourth, weekend predictions produce more false positives, so officers and analysts should apply additional caution when using model outputs in weekend operational planning.

Overall, the model is most useful for **resource prioritization, analytical triage, and patrol planning**, not for determining outcomes in individual cases.

4.3 Limitations and constraints

This study has several limitations and operational constraints.

1. **Limited variable scope:** only location, time, and crime type are included.
2. **Incomplete mapping coverage:** a small number of records were not fully classified by the crime dictionary.
3. **Uneven subgroup sizes:** some spatial categories contain few observations.
4. **Class imbalance sensitivity:** performance depends strongly on balancing strategy.
5. **Predictive rather than causal:** the model identifies associations, not causes.
6. **Not suitable for high-stakes automation:** outputs should support human judgment rather than replace it.

5. Conclusion

This project developed and compared four classification models to predict arrest outcomes using only spatial, temporal, and crime-type information from the Chicago crime dataset. Among the tested models, the **balanced Decision Tree** was selected as the final model because it offers the best overall compromise between **Recall, F1-score, AUC-ROC, and interpretability**.

The analysis shows that **tree-based models outperform the linear baseline**, that **crime type is the strongest predictor**, and that predictive performance varies across both **location** and **time**. Cross-validation and robustness checks further show that the model is reasonably stable but sensitive to class-imbalance handling. The model therefore provides useful support for pattern analysis and operational planning, but it should be used cautiously and never as a fully automated decision system.

Appendix

GitHub repository link:

https://github.com/SaberAlterP/Team15_IT5006_Predicive_Policing_AY2526Sem2

AI Tool Declaration:

Our team used Gemini-3, GPT-5.4 to support the drafting, restructuring, and language polishing of this report. And our team used AI tool to optimize our code. Our team are responsible for the content and quality of the submitted work.